

Machine Learning (WiSe2025-26)



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die Vorstellung



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Teaching Staff for Special Task

SPD:

Applied AI in Digital Production Management (M-MDM)

Examination Board Chairman

Student Advisor – M-MDM | Sensor Lab

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Master (EE)
Ph.D (EE)
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Mumbai,
Indien



Faculty (EE)
IIT, Guwahati
Assam, Indien

Educational Experiences (10 jahre)



Industrial Experiences (16 years)



Consultancy (three years)



Training | Teaching and Area of Interests

AI based concept product development using edge devices

AI with Python and OOP Programming | Lab

Machine learning and Deep learning | Lab

Computer vision with OpenCV

Data Acquisition and Control using LabVIEW

Case Study: Intelligent System

Case Study: AI Project

Case study: Mechatronic System Simulation using Matlab.

Area of Interests

Large language Models (LLM)

Natural language processing (NLP)

Multimodal AI (text, audio, video ,...)

Agentic AI

the principle of similarity where it predicts the label or value of a new data point by considering the labels or values of its K nearest neighbors

K-Nearest Neighbor (KNN) Algorithm - Classification

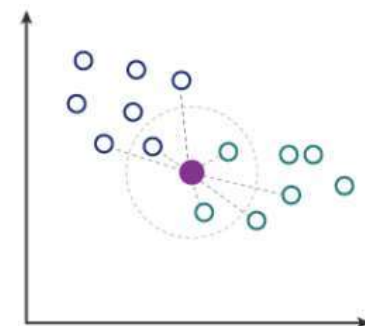
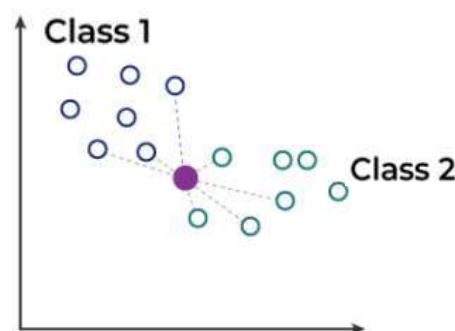
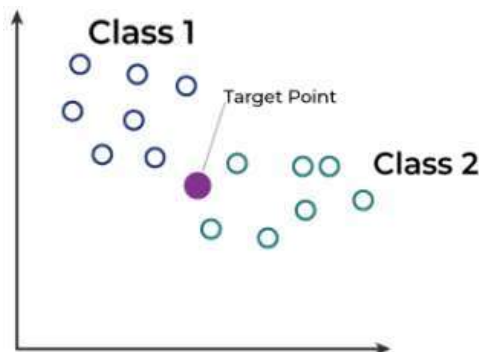
It is a supervised machine learning algorithm generally used for classification

Step 1:

It works by finding the "k" closest data points (neighbors) to a given input

Step 2:

It makes a predictions based on the majority class (for classification) or the average value (for regression).



Properties of KNN:

Simple to use:

Easy to understand and implement.

No training step:

No need to train as it just stores the data and uses it during prediction.

Few parameters:

Only needs to set the number of neighbors (k) and a distance method.

Versatile:

Works for both classification and regression problems.

Slow with large data:

Needs to compare every point during prediction.

Struggles with many features:

Accuracy drops when data has too many features.

Can Overfit:

It can overfit especially when the data is high-dimensional or not clean.

Applications of KNN:

Recommendation Systems:

Suggests items like **movies or products** by finding **users with similar preferences**.

Spam Detection:

Identifies **spam emails** by **comparing new emails** to known **spam and non-spam examples**.

Customer Segmentation:

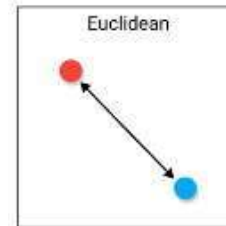
Groups customers by comparing **their shopping behavior** to others.

Speech Recognition:

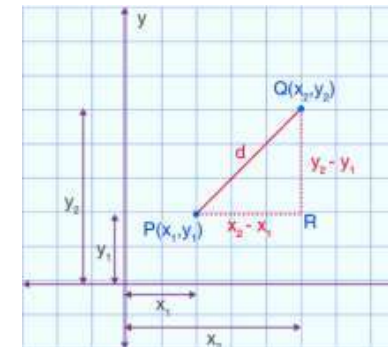
Matches spoken words to **known patterns** to convert them **into text**.

K-Nearest Neighbor (KNN) Predict the Class labels

Sepal length (cm)	Sepal width (cm)	Species
5.3	3.7	Setosa
5.1	3.8	Setosa
7.2	3	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Versicolor
7.3	2.9	Virginica
6	2.7	Veriscolor
5.8	2.8	Virginica
6.3	2.3	Veriscolor
5.1	2.5	Veriscolor
6.3	2.5	Veriscolor
5.5	2.4	Veriscolor



$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



Using Pythagoras theorem,

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

$$PQ^2 = PR^2 + QR^2$$

$$\text{Therefore, } d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Test data

Sepal length (cm)	Sepal width (cm)	Species
5.2	3.1	??

Computing the Euclidean distance

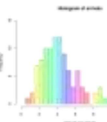
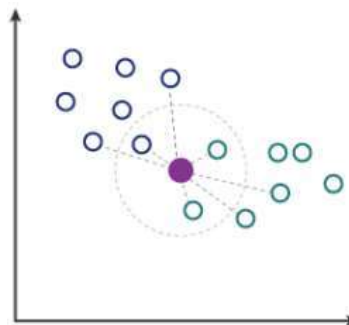
Sepal length (cm)	Sepal width (cm)	Species	Euclidean distance (cm)
5.3	3.7	Setosa	0.61

Distance calculation and Rank Assignment:

Sepal length (cm)	Sepal width (cm)	Species	Euclidean distance (cm)
5.3	3.7	Setosa	0.61
5.1	3.8	Setosa	0.71
7.2	3	Virginica	2.00
5.4	3.4	Setosa	0.36
5.1	3.3	Setosa	0.22
5.4	3.9	Setosa	0.82
7.4	2.8	Veriscicolor	2.22
7.3	2.9	Virginica	2.11
6	2.7	Veriscicolor	0.89
5.8	2.8	Virginica	0.67
6.3	2.3	Veriscicolor	1.36
5.1	2.5	Veriscicolor	0.61
6.3	2.5	Veriscicolor	1.25
5.5	2.4	Veriscicolor	0.76

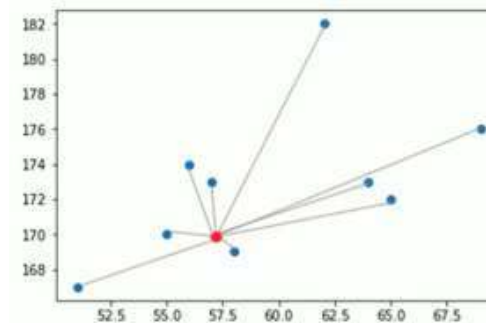
Sort Euclidean distance column in ascending order

Sepal length (cm)	Sepal width (cm)	Species	Euclidean distance (cm)	Rank
5.1	3.3	Setosa	0.22	1
5.4	3.4	Setosa	0.36	2
5.3	3.7	Setosa	0.61	3
5.1	2.5	Veriscicolor	0.61	4
5.8	2.8	Virginica	0.67	5
5.1	3.8	Setosa	0.71	6
5.5	2.4	Veriscicolor	0.76	7
5.4	3.9	Setosa	0.82	8
6	2.7	Veriscicolor	0.89	9
6.3	2.5	Veriscicolor	1.25	10
6.3	2.3	Veriscicolor	1.36	11
7.2	3	Virginica	2.00	12
7.3	2.9	Virginica	2.11	13
				14



Quiz: KNN

Height (cm)	Weight (Kg)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
170	57	???

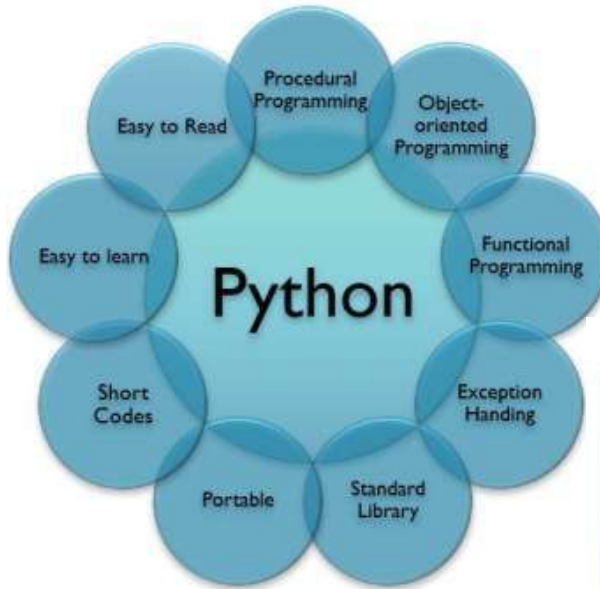


Why make machine learn?

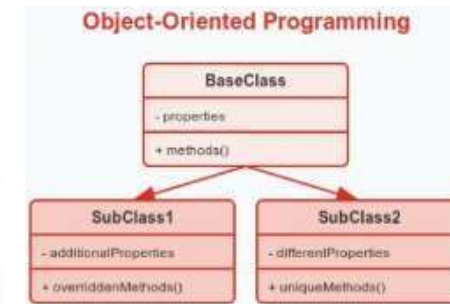
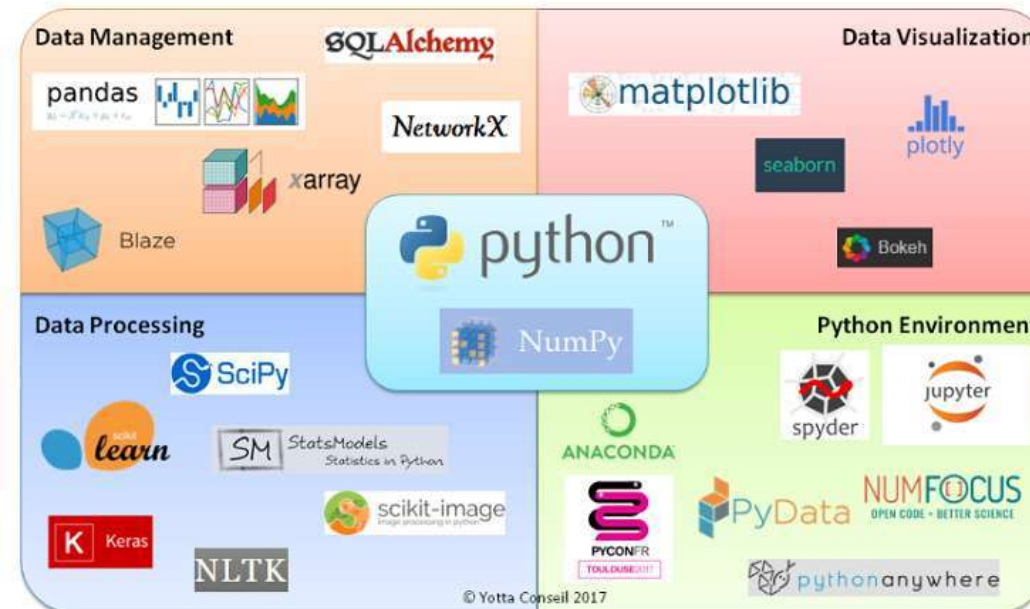
Leverage data to make data-driven decisions at scale using learning algorithms

- Lack of sufficient human expertise in a domain (e.g., simulating navigations in unknown territories)
- Scenarios and behavior can keep changing over time (e.g infrastructure in an organization)
- Humans have sufficient expertise in the domain but it is extremely difficult to formally explain or translate this expertise into computational tasks (e.g., speech recognition, translation, scene recognition, cognitive tasks, and so on).
 - Solution need to adapted to particular case (user biometrics)

Python Eigenschaften :



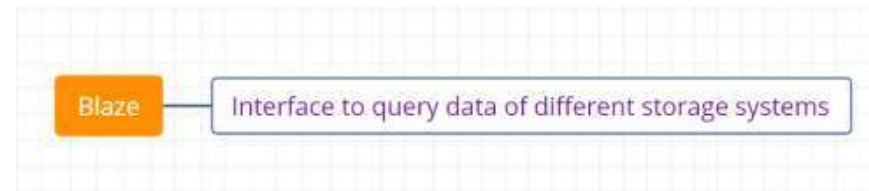
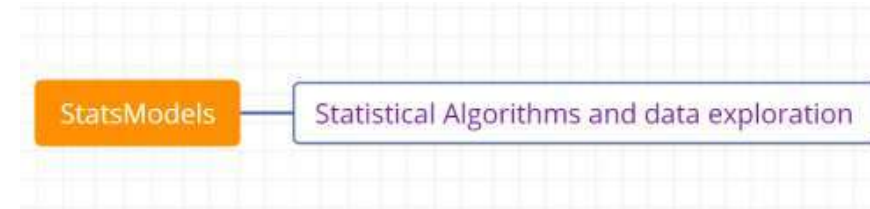
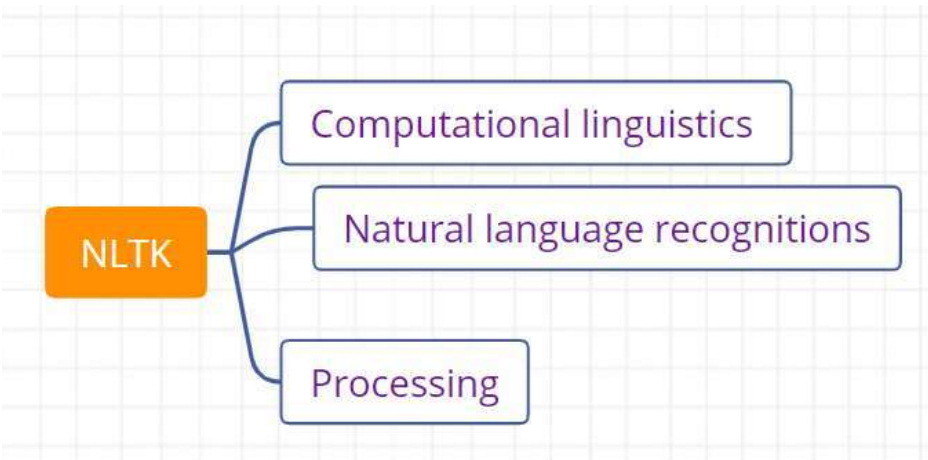
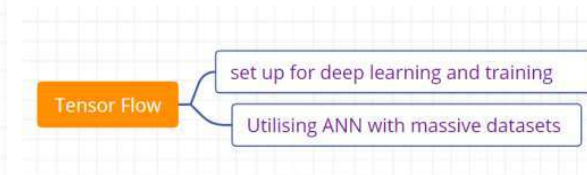
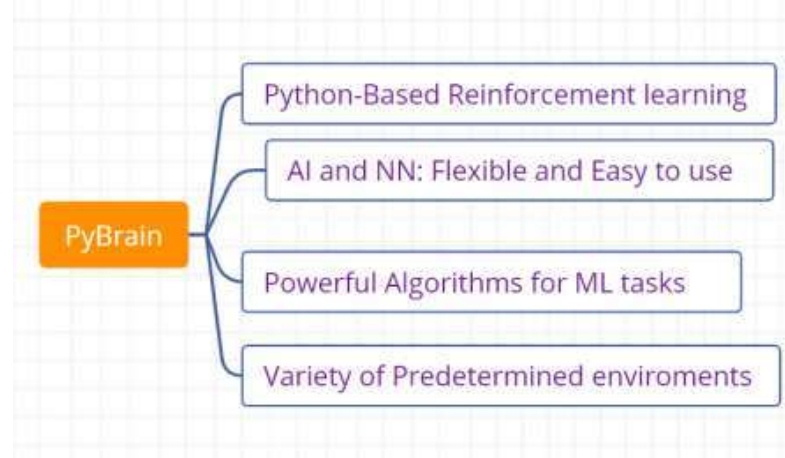
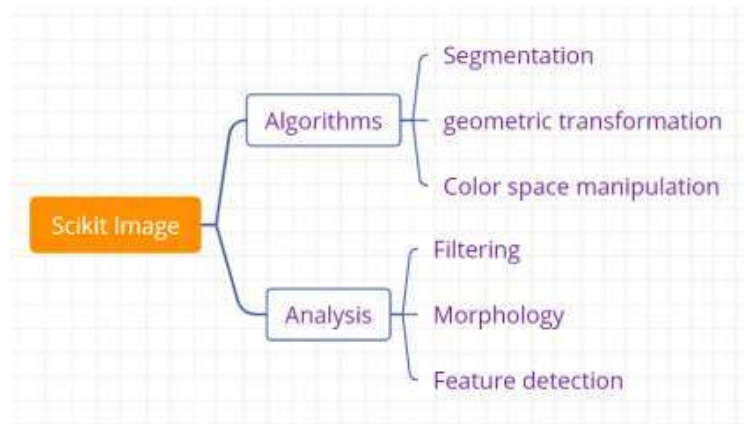
Modularity: When two different systems have **two distinct jobs** at hand but are grouped to perform a bigger job first



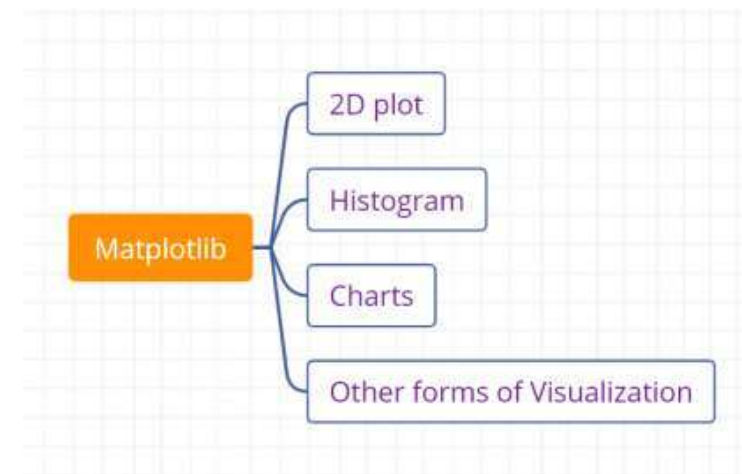
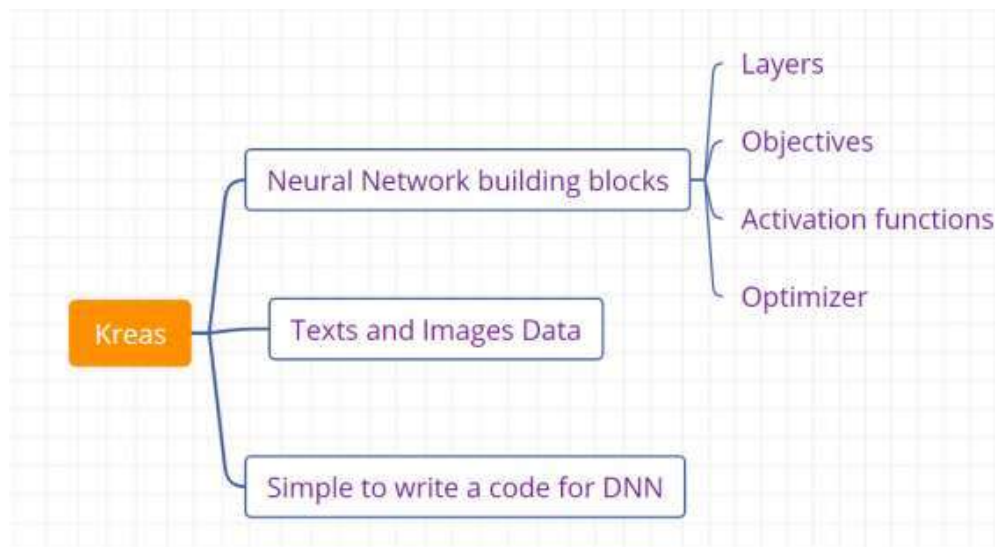
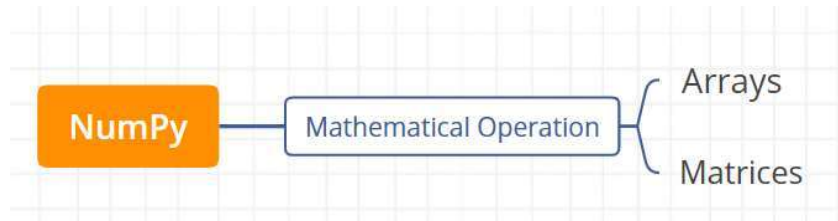
Supermarkt:



Python Packages:



Python Packages:

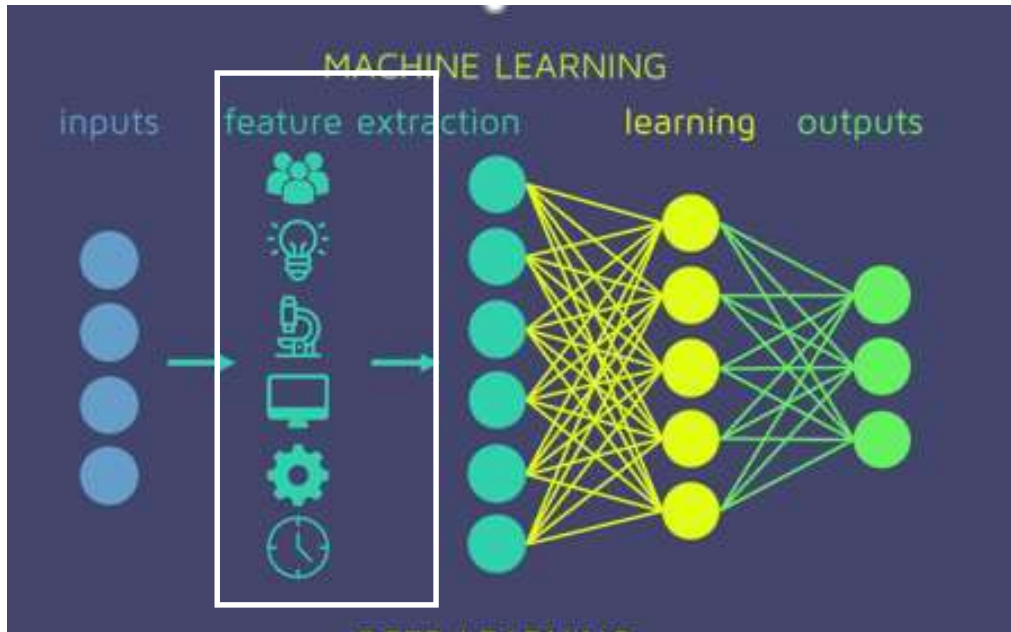


Why make machine learn?

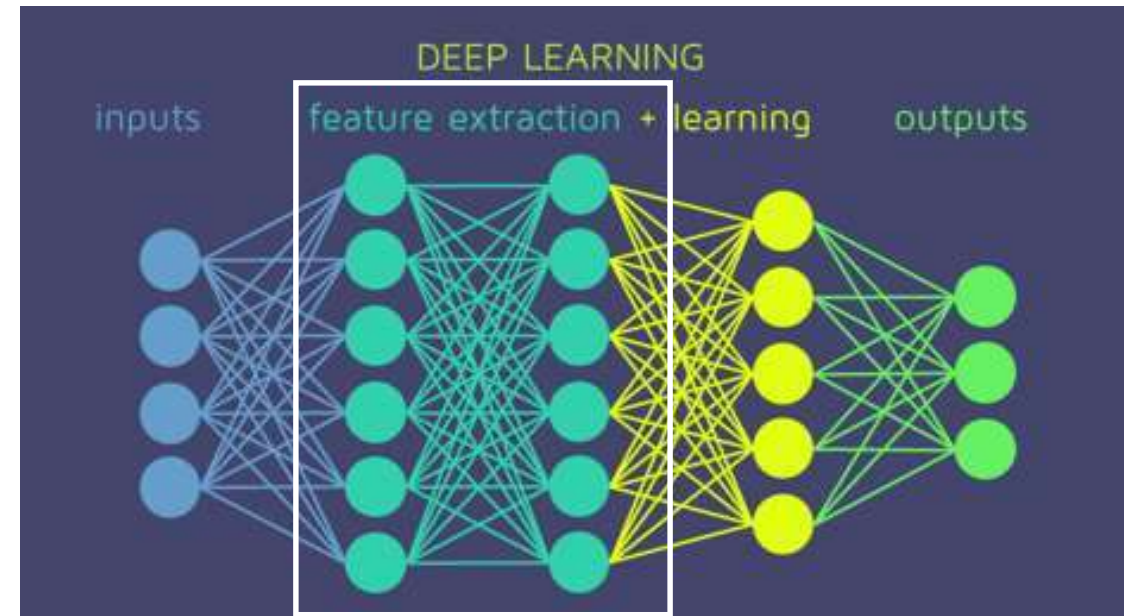
Leverage data to make data-driven decisions at scale using learning algorithms

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Machine learning and Deep learning



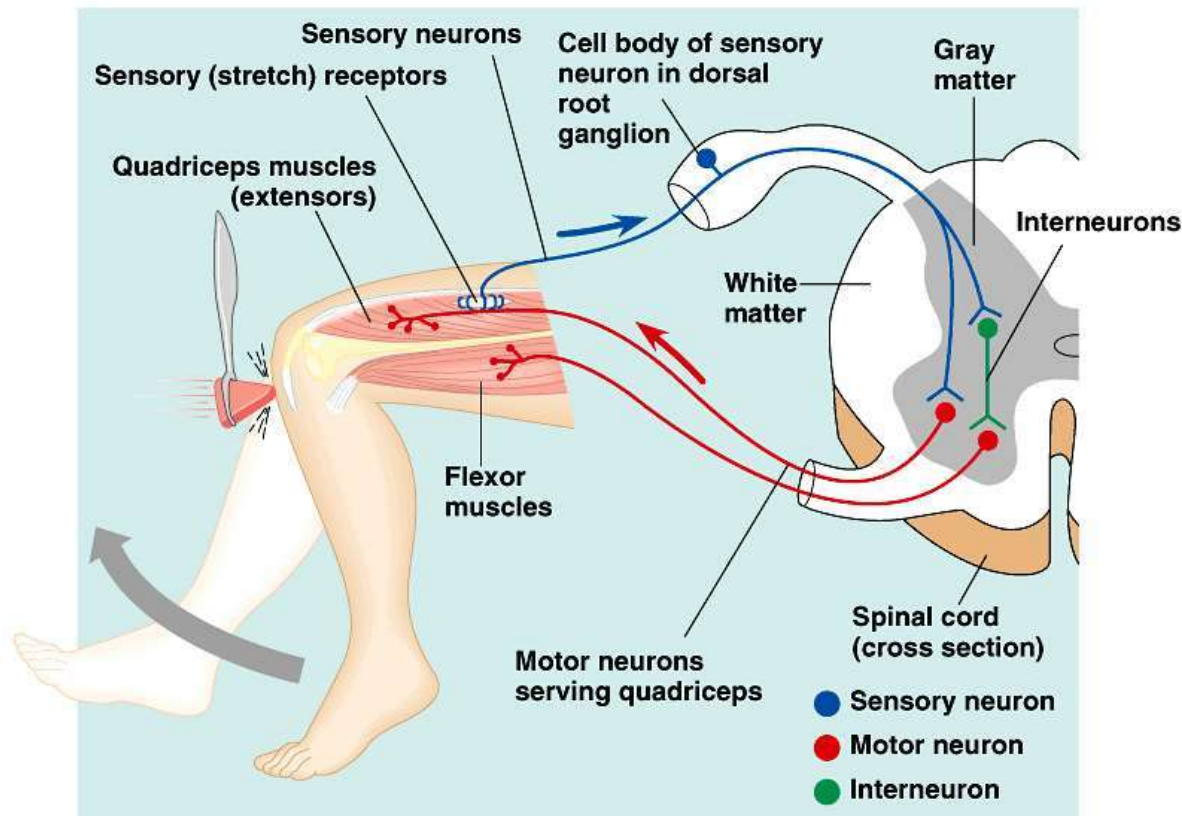
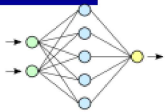
Machine learning



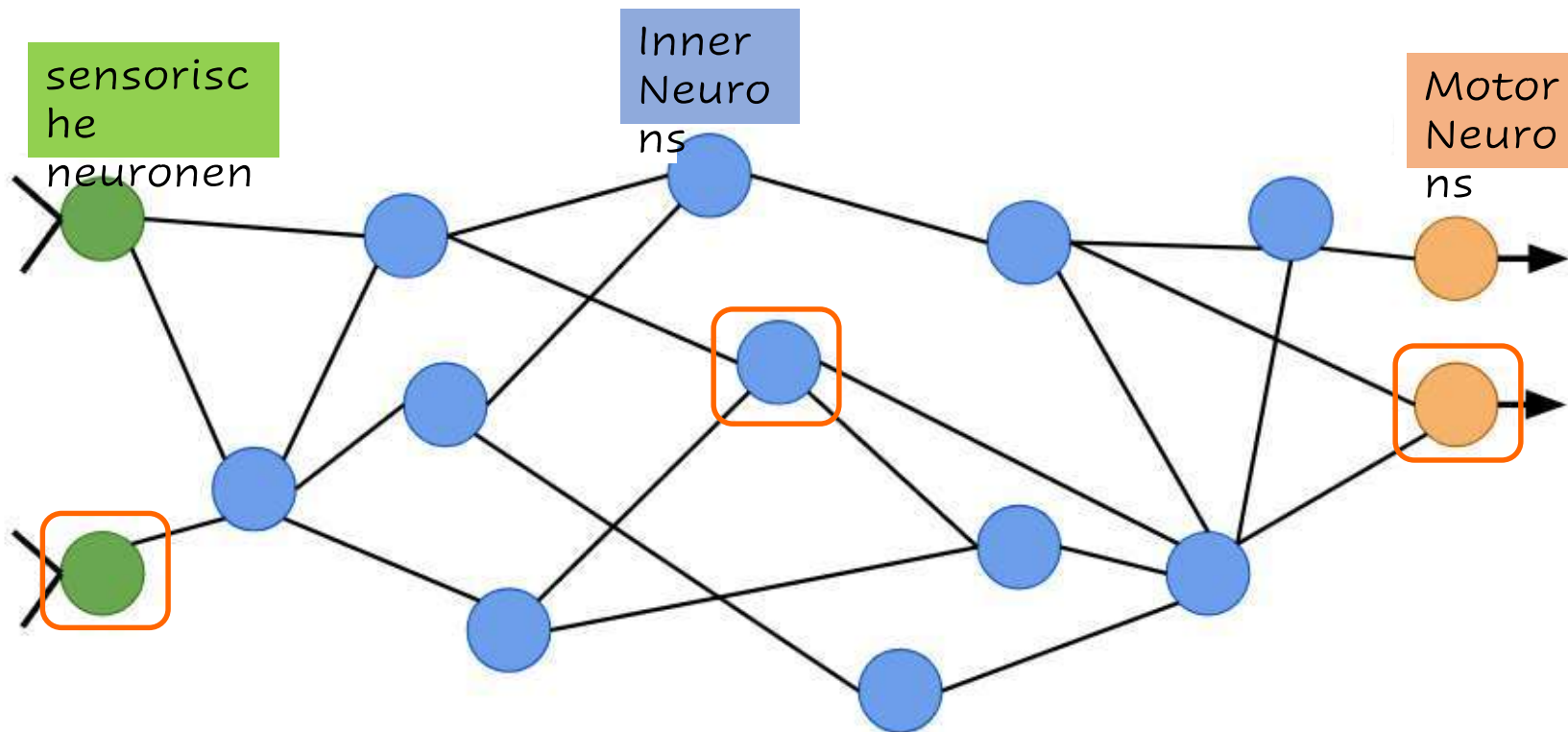
Deep learning

Einfacher Nervenkreislauf: Reflexbogen

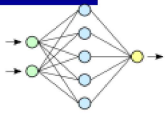
- Ein Reflex ist eine autonome Reaktion.



Einfacher Nervenkreislauf: Reflexbogen



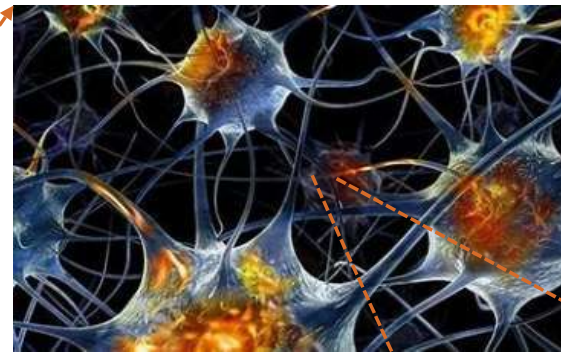
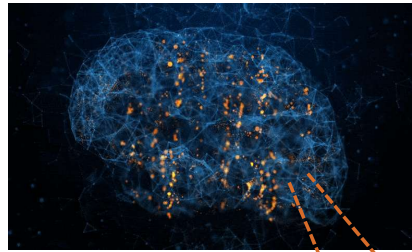
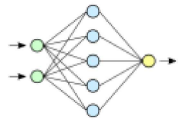
Neuronenfeuerungsmuster:



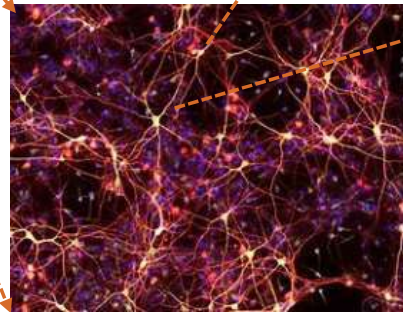
Zum Speichern und Wiedergeben

*© N974 study suggests our brains use
distinct firing patterns to store and replay
memories*

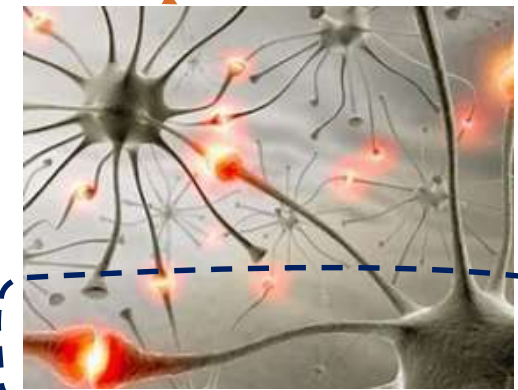
Neuronales Netzwerk: Biologisches Neuron



Neuronen



Neuronales Netzwerk

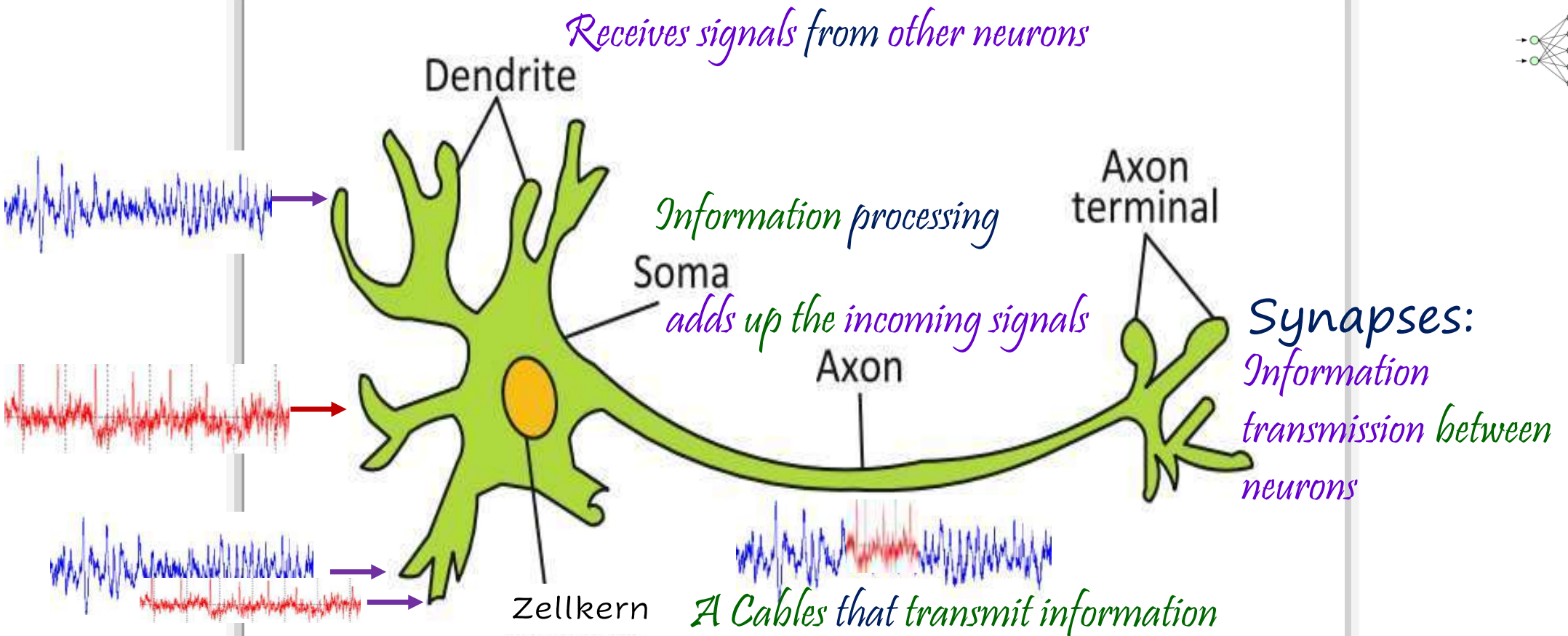


Biologisches Neuron

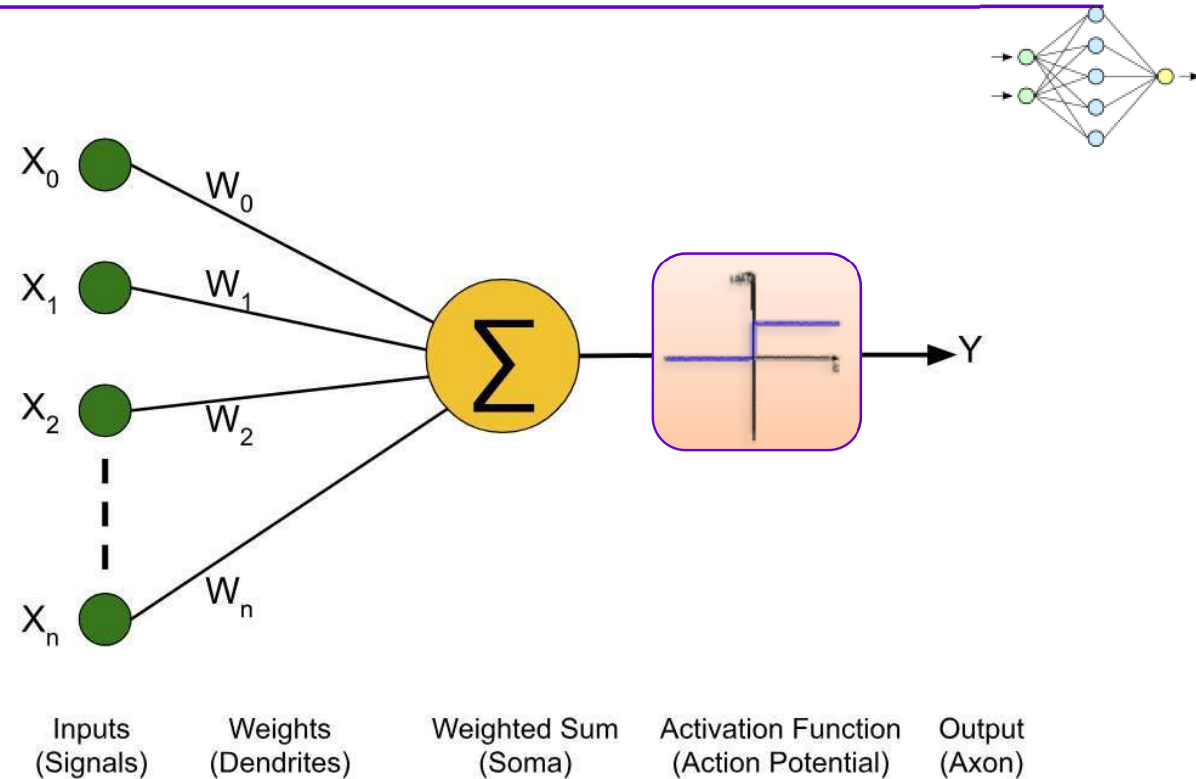
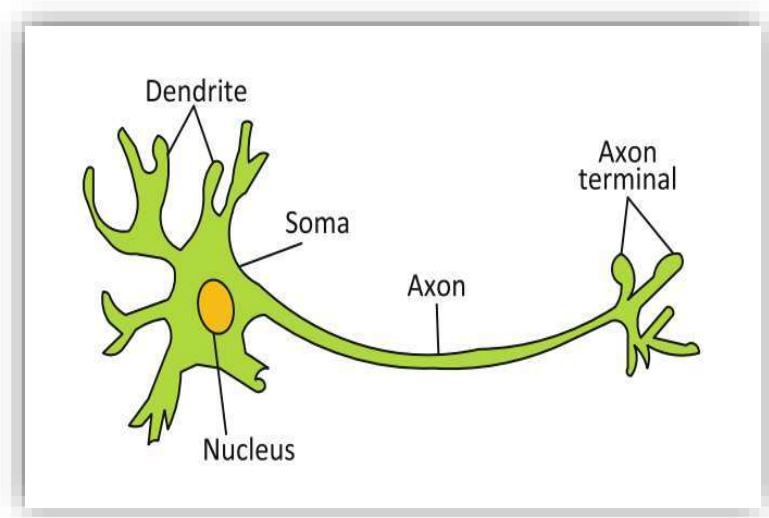


Biological neuron

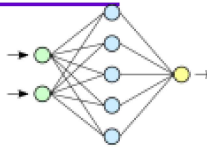
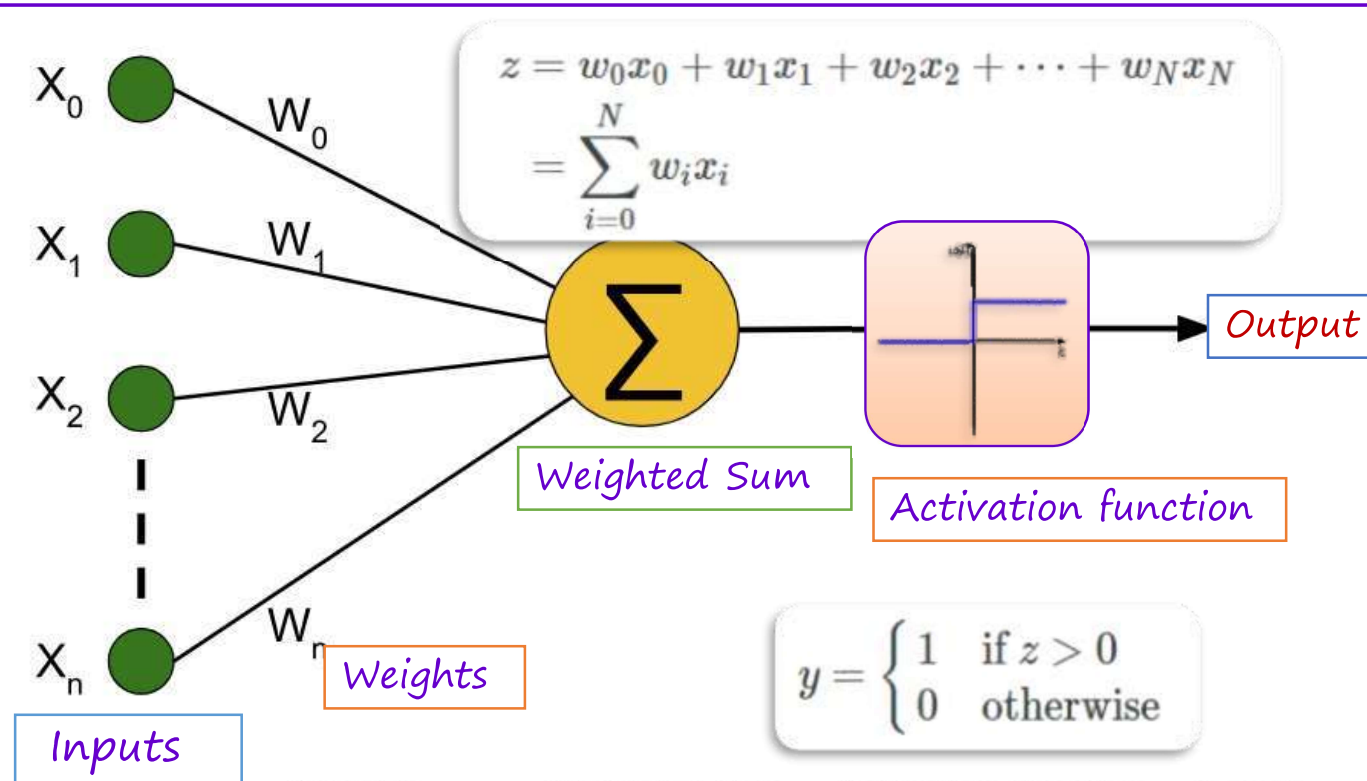
A special biological cell that
processes information



Neuron Mathematica Model



Neuron Mathematica Model

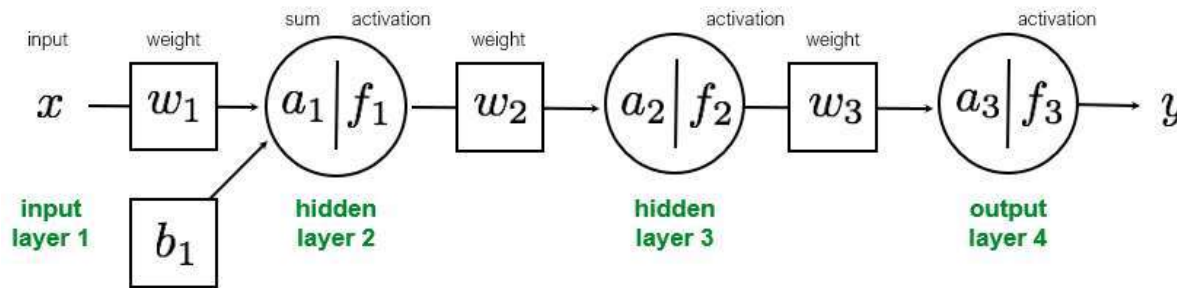


BNN and ANN:

Similarities:	
Biological Neural Network	Artificial Neural Network
Soma	Node
Dendrites	Input
Synapse	Weights or Interconnections
Axon	Output
Differences	
Massively parallel, slow but superior than ANN	Massively parallel, fast but inferior than BNN
10^{11} neurons and 10^{15} interconnections	10^2 to 10^4 nodes Mainly depend upon type of application
They can tolerate ambiguity	Very precise, structured and formatted data is required to tolerate ambiguity
Performance degrades with even partial damage	It is capable of robust performance, hence has the potential to be fault tolerant
Stores the information in the synapse	Stores the information in continuous memory locations



Forward feed propagation (FFP)



$$a_1 = w_1 \cdot x + b_1$$

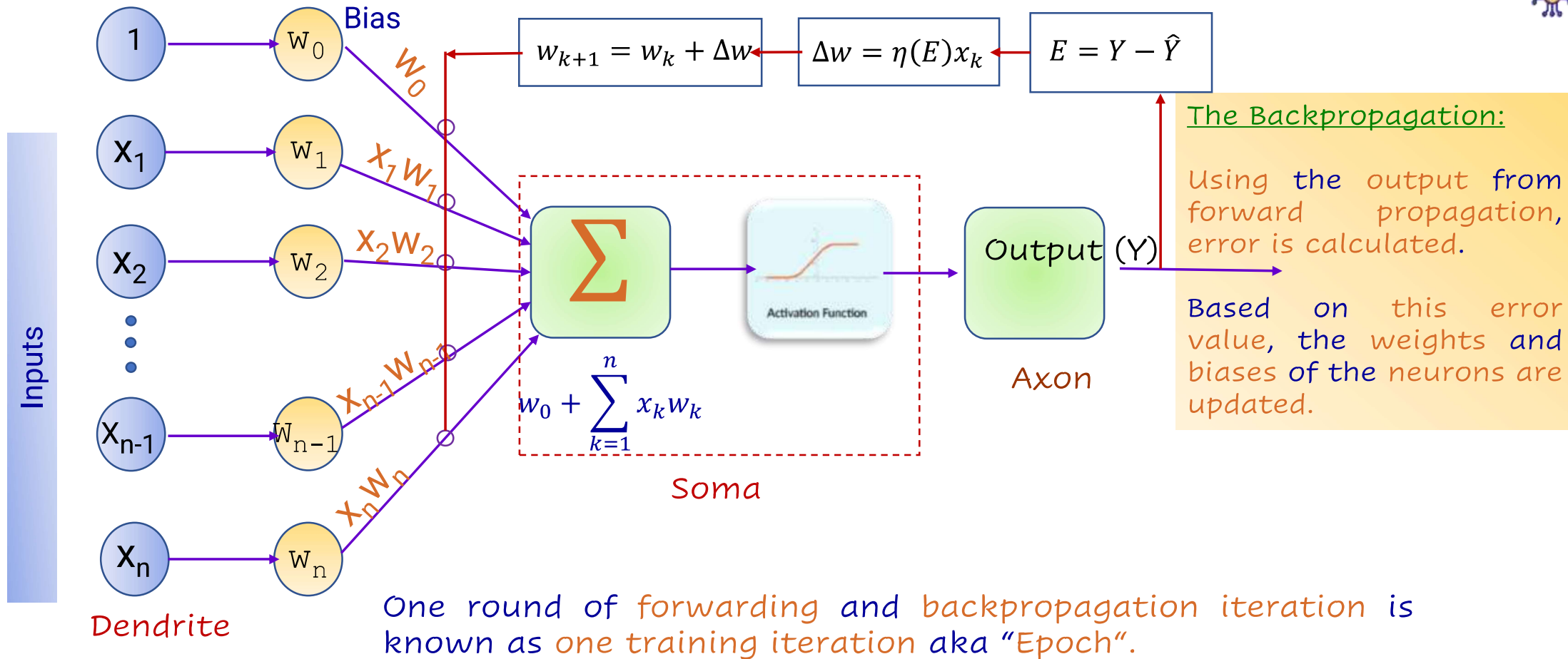
$$a_2 = w_2 \cdot f_1(w_1 \cdot x + b_1)$$

$$a_3 = w_3 \cdot f_2(w_2 \cdot f_1(w_1 \cdot x + b_1))$$

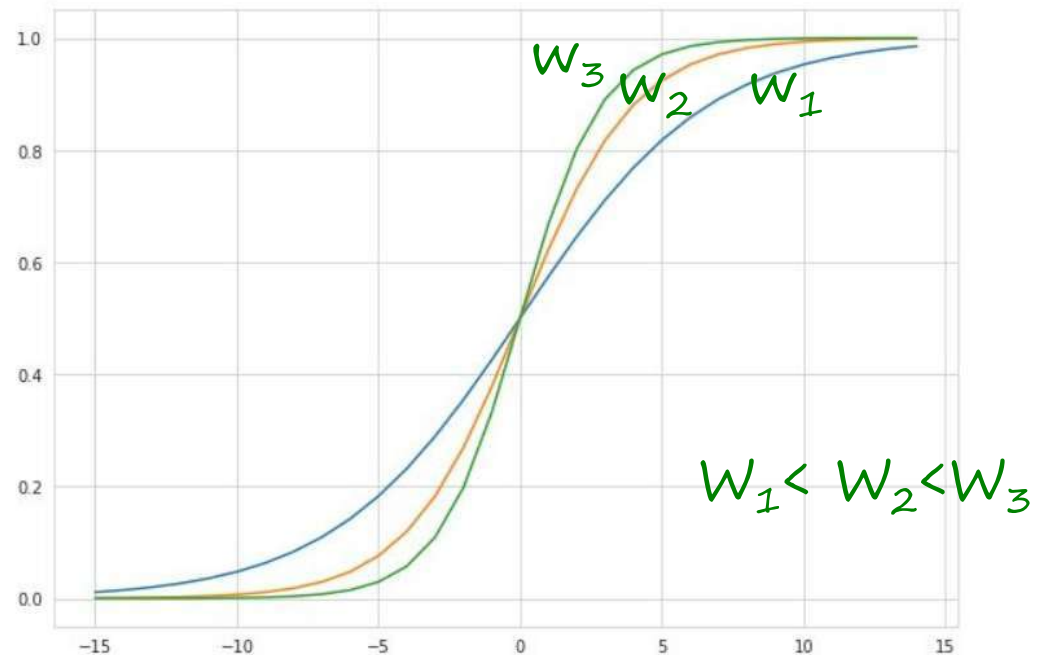
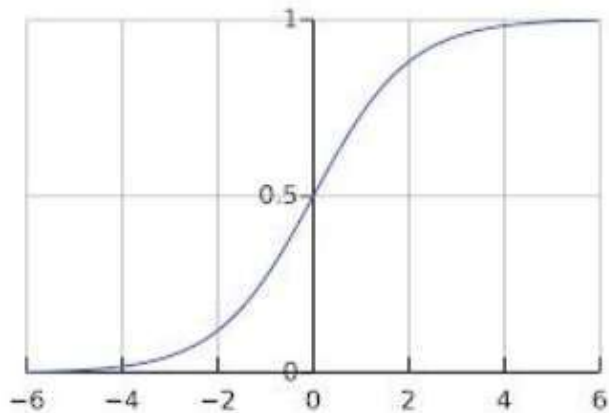
$$y = f_3(w_3 \cdot f_2(w_2 \cdot f_1(w_1 \cdot x + b_1)))$$



Backpropagation and Epoch

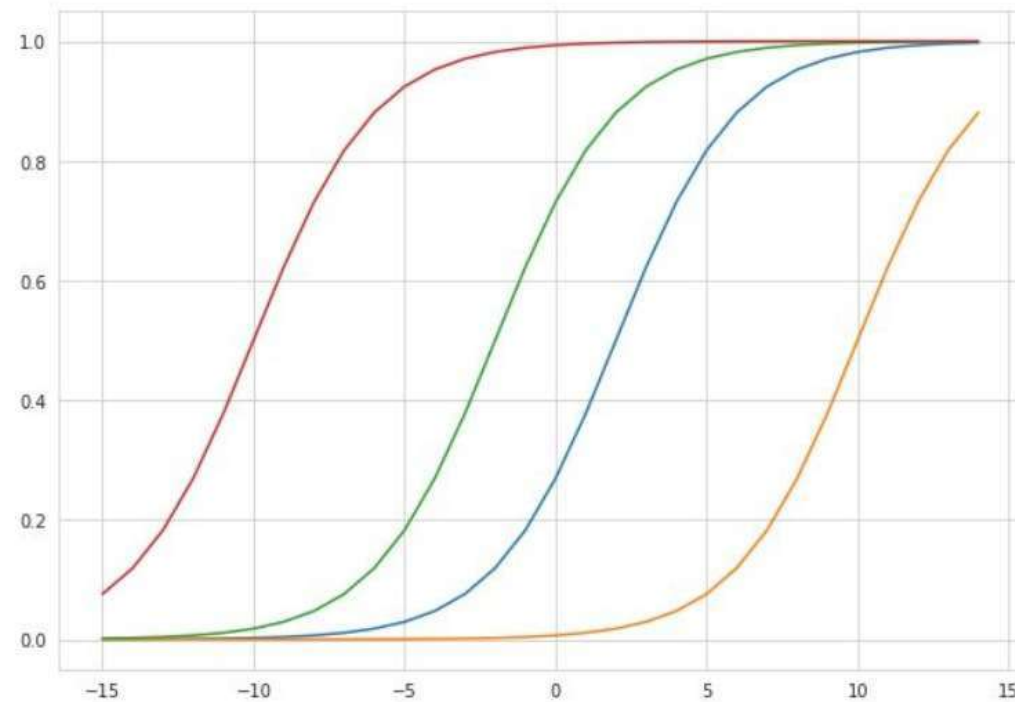
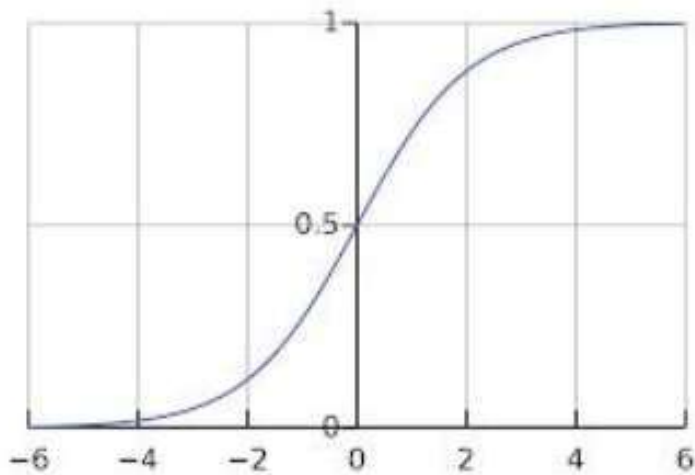


Sigmoid function: Weight variation (bias = constant)



the **steepness** of the curve will change.

Sigmoid function: bias variation (Weight = constant)



bias is required for *shifting the origin* of the curve to the *left* or *right*.

Training dataset too large to be feed in computer at once.

Batch, Batch size, and Iterations

Batch:

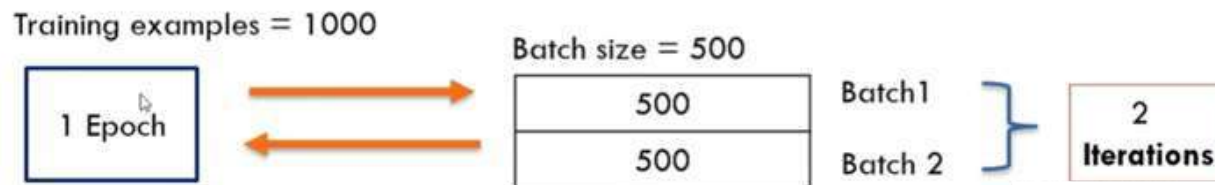
To divide the training dataset into multiple smaller trainingsets.

Batch size:

Total number of training examples present in a single batch.
Batch size is a hyperparameter.

Iterations:

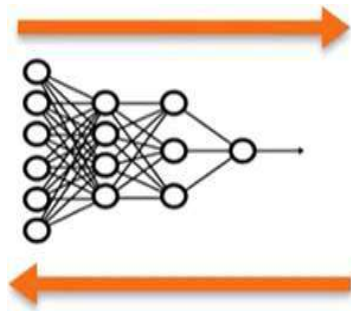
Iterations are the number of batches needed to complete one epoch.



Epoch:

One entire passing of training data through the algorithm - Epoch

One forward pass and one backward pass of all the training examples.

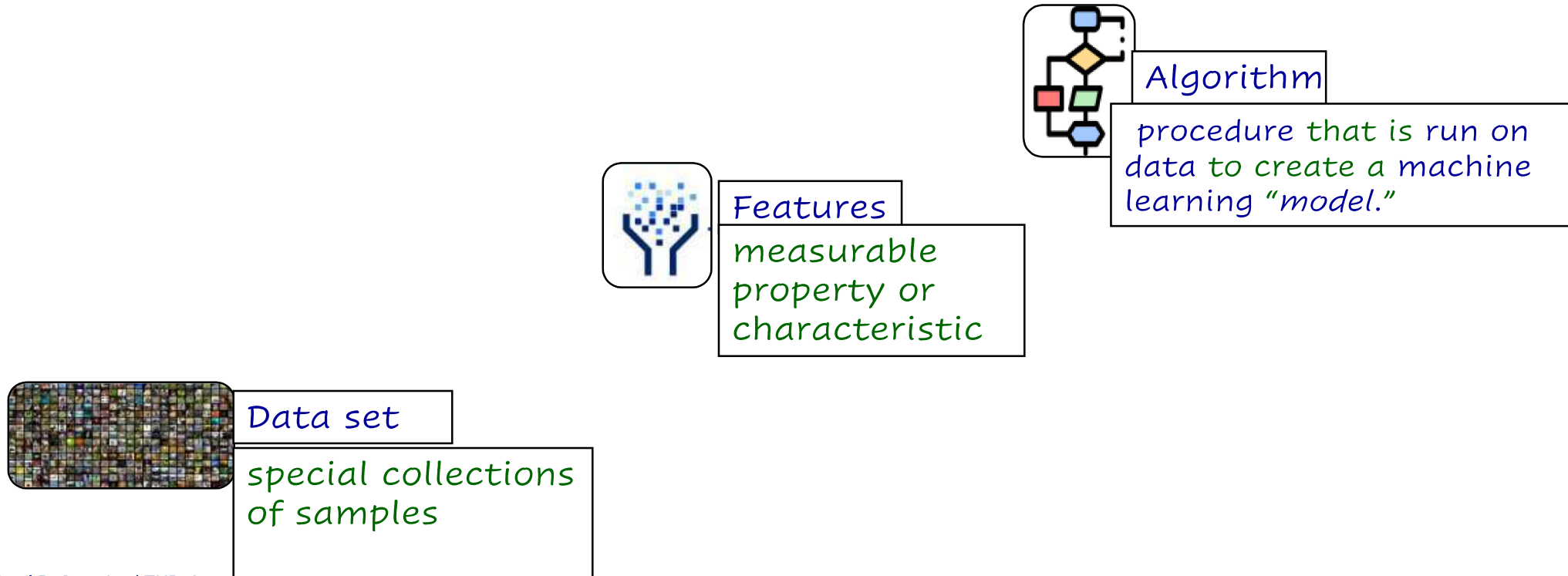


1 Epoch The weight will be decent
2 Epoch
3 Epoch
⋮
N Epoch The weight will be improved

ML Components

machine learning = learning and inference
*a conclusion reached on the basis
of evidence and reasoning.*

Machine Learning



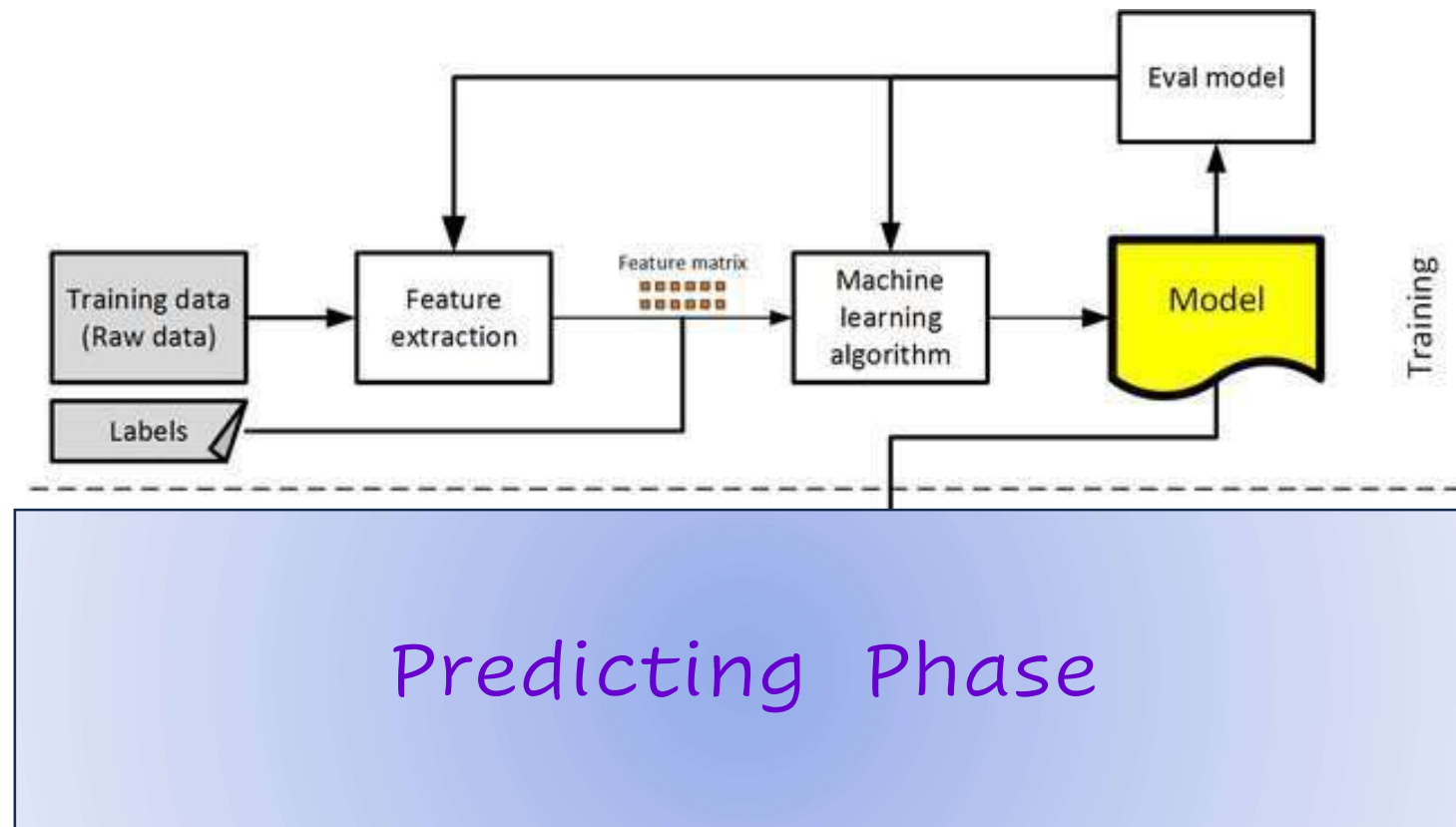
ML: Training and Predicting



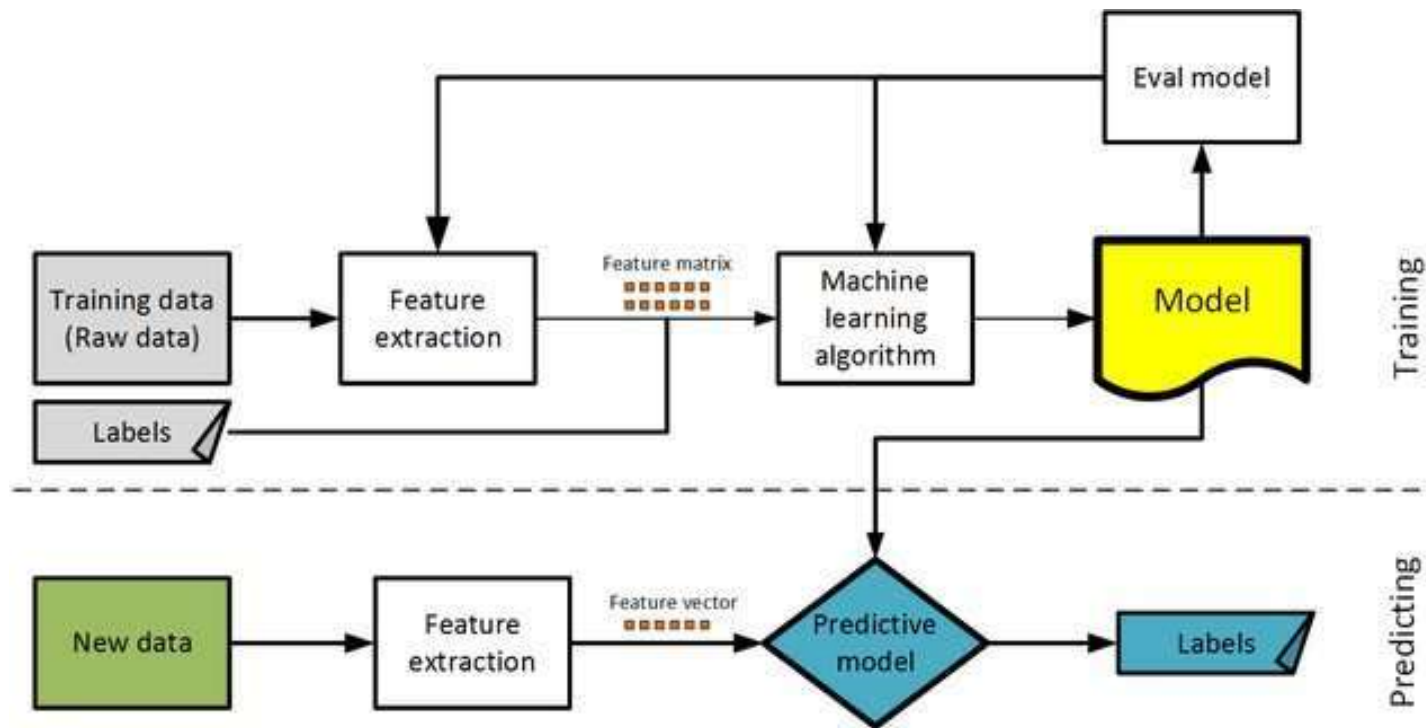
Training Phase

Predicting Phase

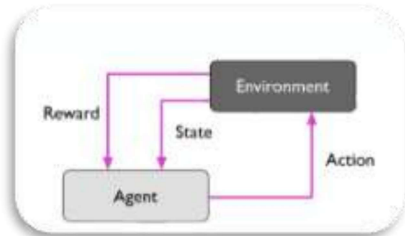
ML: Training and Predicting



ML: Training and Predicting

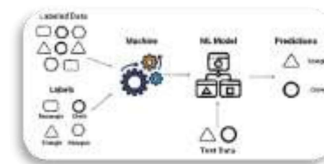


ML Algorithms Classification:

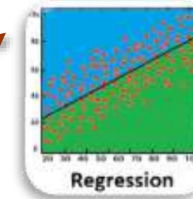


Reinforcement Learning

Robot navigation
Learning tasks
Real time decisions



Supervised Learning



Market forecasting
Advertising popularity prediction
Estimating life expectancy

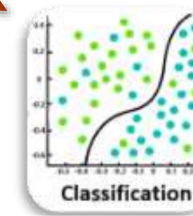
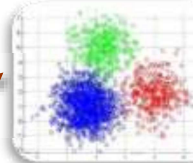
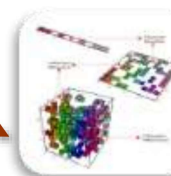


Image classification
Diagnostics
Customer retention



Customer segmentation
Target marketing
Recommender system

Clustering

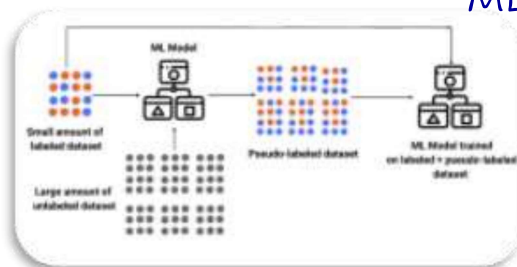


Big data visualization
Meaningful compression
Feature Elicitaion

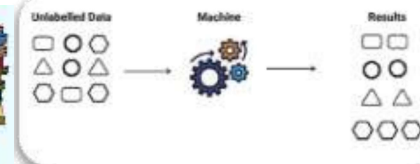
Dimensionality reduction



ML Algorithms Classification



Semi-Supervised Learning

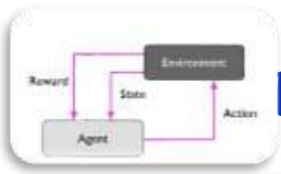


Unsupervised Learning

Elicitation techniques are a category of research tasks that use visual, verbal, or written stimuli to encourage participants to talk about their ideas



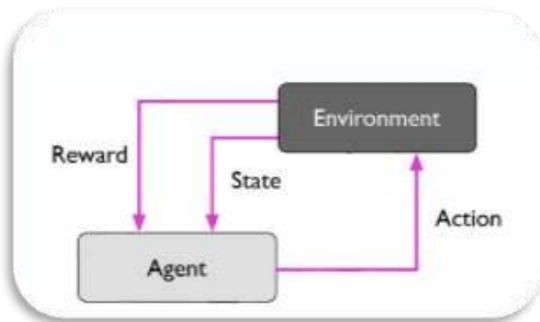
RL Example



PacMan game

Initial Release date: May 22, 1980.

It is a maze action video game developed and released by Namco



The Objective:

You control the yellow, circular character through a maze, eating all **Pac-Dots** while avoiding four ghosts.

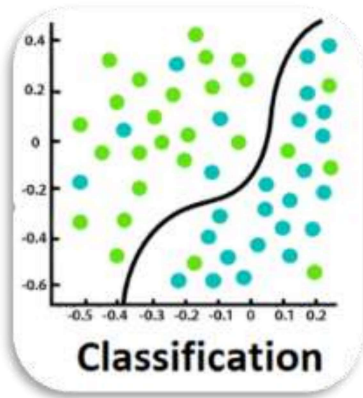
Let's take the game of PacMan where the goal of the agent (PacMan) is to eat the food in the grid while avoiding the ghosts on its way.

The grid world is the interactive environment for the agent.

PacMan receives a reward for eating food and punishment if it gets killed by the ghost (loses the game).

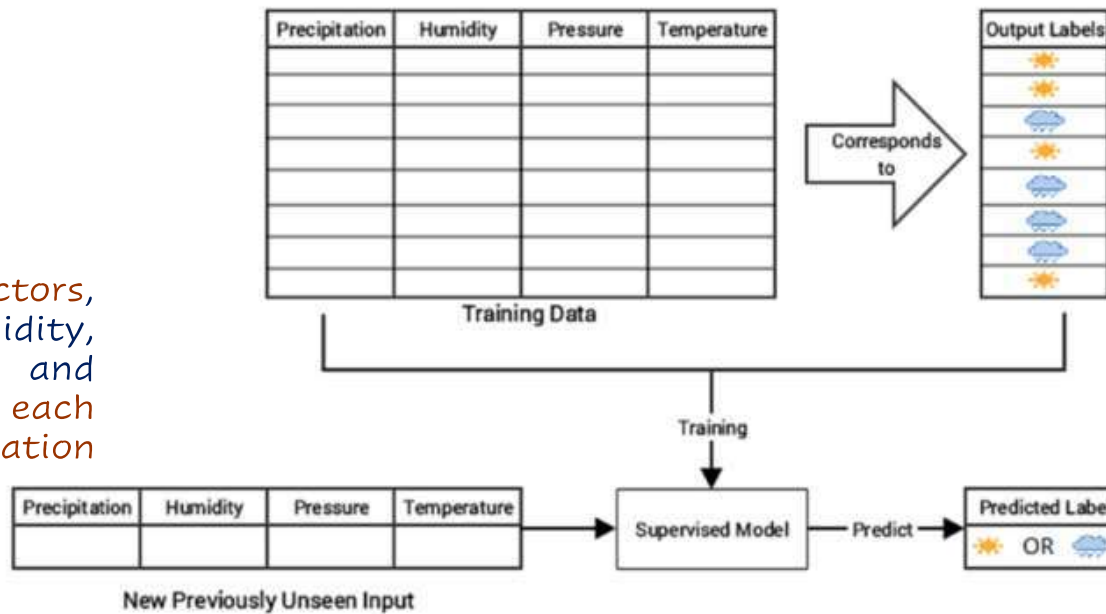
The states are the location of PacMan in the grid world and the total cumulative reward is PacMan winning the game.

Classifications



The key **objective** is to **predict output labels** or responses that are **categorical in nature** for **input data**

feature vectors,
(precipitation, humidity,
pressure, and
temperature) for **each**
data sample/observation

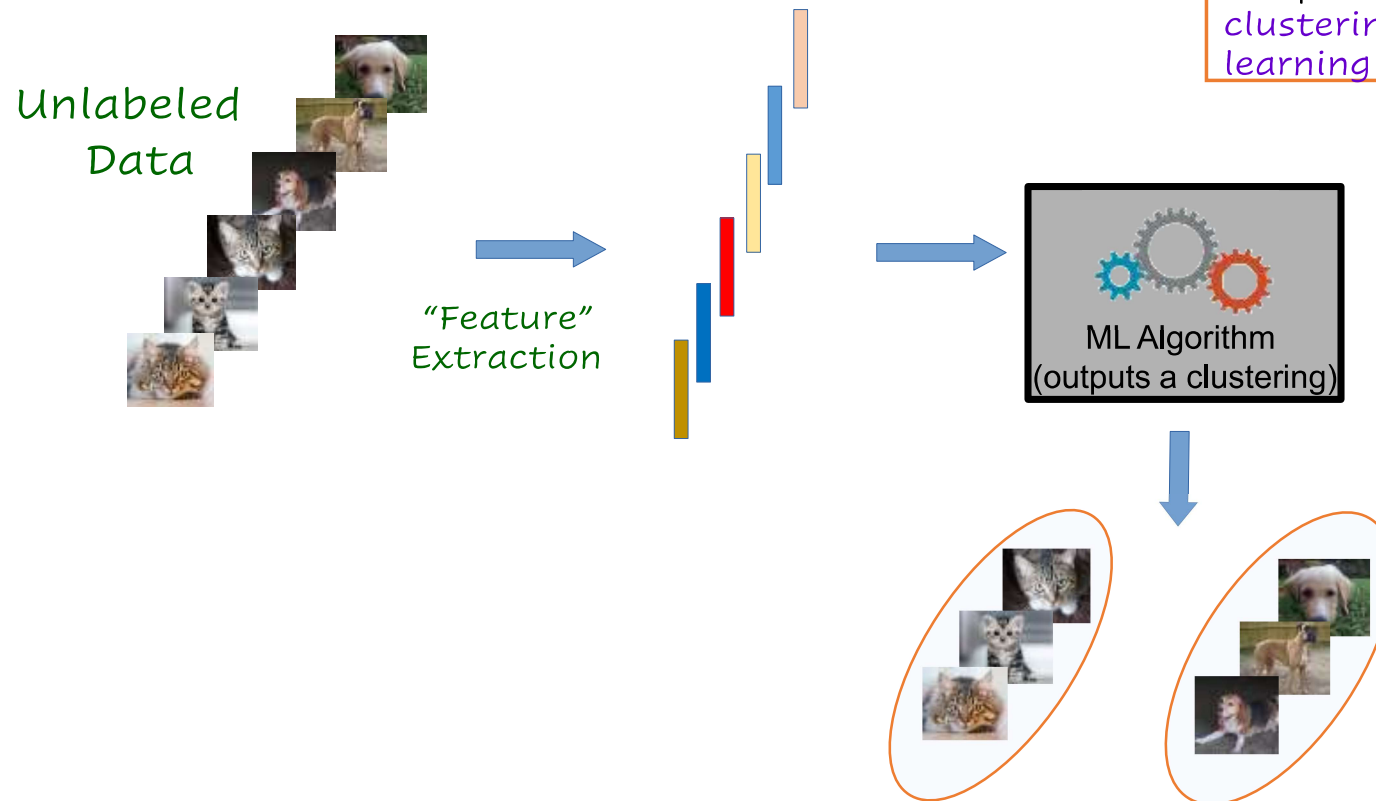


Popular classification algorithms include **logistic regression**, **support vector machines**, **K-nearest neighbors**, and many more

a real-world example of predicting the weather.



Unsupervised Learning workflow



Note: This example is for the problem of *data clustering*, an unsupervised learning problem

